MIL-P-36705A
13 December 1985

SUPERSEDING
MIL-P-36705
19 August 1970

MILITARY SPECIFICATION

PULP PROTECTOR, DENTAL, ZINC OXIDE AND EUGENOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a zinc oxide and eugenol paste suitable as a hard-setting pulp protector and cavity base.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

MMM-A-250	- Adhesive, Water-Resistant, (For Closure of Fiberboard Boxes).
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.
PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-T-76	- Tape, Packaging, Paper (For Carton Sealing).

AMSC N/A FSC 6520

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

U. S. PHARMACEUTICAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

AMERICAN DENTAL ASSOCIATION (ADA)

Council on Dental Materials and Devices Accepted Dental Therapeutics.

(Application for copies should be addressed to the Council on Dental Materials and Devices, American Dental Association, 211 East Chicago Avenue, Chicago, IL 60611.)

U. S. DEPARTMENT OF HEALTH AND HUMAN RESOURCES,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

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(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The paste shall consist essentially of a zinc oxide base paste and a eugenol accelerator paste. All ingredients entering into the formulations listed in 3.1.1 and 3.1.2 shall be of The United States Pharmacopeia (USP) or National Formulary (NF) quality, or if not included in either of these compendia, shall be of quality suitable for pharmaceutical use. The pulp protector shall be certified as acceptable by the Council on Dental Materials and Devices of the American Dental Association. A mixing pad made of specially treated paper suitable for mixing the pastes together shall also be furnished.

3.1.1 Base paste. The zinc oxide base paste shall contain the following percentages of ingredients by formula weight:

Zinc Oxide (USP)	85 percent
Mineral Oil (USP)	12 percent
Additives	3 percent

The additives present shall not detract from the serviceability of the pulp protector and shall not be injurious to the mouth. Percentage and presence of ingredients shall be ascertained from the manufacturer's batch formula control records.

3.1.2 Accelerator paste. The eugenol accelerator paste shall contain the following percentages of ingredients by formula weight:

Polymerized Resin	60 percent
Eugenol (USP)	30 percent
Mineral Oil (USP)	7.5 percent
Additives	2.5 percent

The additives present shall not detract from the serviceability of the pulp protector and shall not be injurious to the mouth. Percentage and presence of ingredients shall be ascertained from the manufacturer's batch formula control records.

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3.1.3 Mixing pad. Paper for the mixing pad shall be 35-pound, No. 946 parchmentized Kraft which shall be bleached, fiberless, and lint free. Paper shall be Food and Drug Administration (FDA) approved for contact with food products. Surface of mixing pad shall measure 1-3/8 inches minimum in length and 1-3/8 inches minimum in width. The number of sheets shall be a minimum of 25.

3.2 Design and description. The zinc oxide and eugenol pastes, when combined in accordance with instructions accompanying the unit package, shall be suitable for use as a protector of vital pulp, as a base for amalgam restorations, or as a temporary filling material during dental procedures. The following components shall be present in each unit of issue of the pulp protector:

One tube of paste containing a net weight of 19.0 to 21.0 grams of base paste.

One tube of accelerator paste containing a net weight of 7.0 to 9.0 grams of accelerator paste.

One mixing pad.

One set of instructions for mixing and use.

3.2.1 Immediate containers (base and accelerator pastes). Tubes for base and accelerator pastes shall conform to PPP-C-186, for group B, class 1, type a, closure D tubes. Tube containing the accelerator paste shall be provided with a plastic (teflon) diaphragm measuring 0.609 ± 0.005 inch in diameter and 0.020 ± 0.005 inch in thickness. Diaphragm shall have a self-closing center slit to provide accurate control of the amount of paste dispensed. A cold seal shall be applied to the inner tube wall at the crimp end (of the accelerator paste tube) to decrease the possibility of leakage. Tubes shall be permanently and legibly marked to permit identification of base and accelerator. Marking shall also include the manufacturer's name, address, batch or lot number of the paste, and date of manufacture.

3.3 Physical properties.

3.3.1 Odor and flavor. The material shall have no unpleasant odor or flavor, when tested as specified in 4.4.2.

3.3.2 Compressive strength. The compressive strength of the mixed and set paste shall be a minimum of 200 pounds per square inch, when tested as specified in 4.4.7.

3.3.3 Properties of mixed paste. Pulp protector shall show high fluidity, shall have a tacky consistency to facilitate adaption of cavity wall; shall be homogeneous, free from lumps and granules; and shall flow smoothly in order that no pressure shall be required during placement, when tested as specified in 4.4.3.

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3.3.4 Setting time. Pulp protector shall have a setting time of 3-3/4 to 4-1/4 minutes, when tested as specified in 4.4.4.

3.3.5 Flow. The flow for the base only shall be 34 ± 3 millimeters (mm) in diameter; the flow for the mixed paste shall be 59 ± 3 mm in diameter, when tested as specified in 4.4.5.

3.3.6 Cracking time. The cracking time for the pulp protector shall be 1-1/2 ± 1/2 minutes after final set, when tested as specified in 4.4.6.

3.4 Delivery. No more than 3 months of time shall have elapsed from date of manufacture of the pastes to date of delivery to the Government.

3.5 Instructions. Adequate and accurate instructions for mixing and use shall accompany each package.

3.6 Lot or control number. The pulp protector shall be marked with the lot or control number.

3.7 Workmanship. The product and its containers shall be free from defects which may detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the name of the ingredient, the lot number, the date of testing, the test method, individual test results, and the specification requirements.

4.1.4 Manufacturer's records. The percentage of ingredients shall be ascertained from the manufacturer's batch formula control number.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQLs) shall be as specified in table I. Unit of product for inspection purposes shall be the unit of issue (one package) of pulp protector.

TABLE I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major	S-2	1.5
Minor	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with table II. The unit of product for test purposes shall be one pulp protector. Where feasible, the same sample unit shall be used for the determination of two or more different test characteristics.

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TABLE II. Sampling for tests.

Characteristic	Requirement	Test procedure	Unit of product to be tested per lot or batch number	Number of determinations per unit of product	Total number of determinations
Odor and flavor	3.3.1	4.4.2	1	1 a/	1
Properties of mixed paste	3.3.3	4.4.3	1	1 a/	1
Setting time	3.3.4	4.4.4	1	1 b/	3
Flow	3.3.5	4.4.5	1	3 b/	3
Cracking time	3.3.6	4.4.6	1	3 b/	3
Compressive strength	3.3.2	4.4.7	1	5 b/	5

a/ A lot shall be rejected if any determination fails the test requirements.

b/ A lot shall be rejected if the average of the determinations per unit of product fails the test.

4.3 Examination. Pulp protector shall be examined for defects including but not limited to those listed in table III.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Any component (base paste, accelerator paste, mixing pad, or instructions) missing in unit package.
102	Instructions not present in unit container, incorrect, incomplete, or illegible.
103	Tubes not legibly marked to identify contents.
104	Accelerator paste tube not provided with a self-closing plastic diaphragm.
105	Net weights of base paste and accelerator paste in tubes not as specified.
Minor	
201	Paper for mixing pad not made of bleached, fiberglass, lint free, parchmentized, Kraft paper.
202	Size or number of sheets in mixing pad not as specified.

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4.4 Tests.

4.4.1 Standard testing conditions. Unless otherwise specified, all physical tests shall be performed under uniform atmospheric conditions of 75 ± 2 degrees Fahrenheit (^{o}F) and 40 ± 10 percent relative humidity. Equipment and material shall be conditioned in the testing room for no less than 10 hours prior to making the tests. Test sample for all tests shall consist of 2.00 grams of base and 1.00 gram of accelerator.

4.4.2 Odor and flavor. The pleasantness or unpleasantness of the odor and flavor shall be determined by smelling and tasting the pulp protector.

4.4.3 Properties of mixed paste. A mix shall be made using the quantities of base and accelerator pastes given in 4.4.1. Pastes shall be mixed for 10 to 20 seconds on the mixing pad using a stainless steel spatula. The mix shall be observed for the properties specified in 3.3.3.

4.4.4 Setting time. Quantities of the base and accelerator pastes specified in 4.4.1 shall be mixed on the mixing pad using a stainless steel spatula. Tests shall be carried out in an atmosphere having an ambient temperature of 75 ± 1°F with a 30 to 40 percent relative humidity (set is accelerated by increased temperature or humidity, and retarded by decreased temperature or humidity). A 10 gauge needle weighing 10.0 grams shall be used. Needle shape shall be an inverse truncated cone ground to a 1/16 inch flat tip. Readings shall be taken every 1/4 minute. The setting time is the total elapsed time from the start of the mix to the time when the paste no longer "pulls up" with the needle.

4.4.5 Flow. At 1-1/2 minutes after start of mix, a calibrated tube shall be used to measure 0.5 cubic centimeters (cm^3) of a standard mix (see 4.4.1). This sample shall then be deposited on a 4 inch by 4 inch glass plate which has been covered with cellophane sheet. The sample shall then be covered with cellophane, a 4 inch by 4 inch glass plate, and a 500 gram weight, and allowed to stand for 10 minutes. At the end of this time, the weight and glass plate shall be removed. The sample shall then be measured across at least two diameters, with a millimeter scale, and the average reading shall be recorded.

4.4.6 Cracking time. Cracking time shall be determined on a mix as specified in 4.4.1. Base accelerator pastes shall be mixed on the mixing pad using a stainless steel spatula such that the mix in the form of a patty occupies about 3/4 of the surface area of the pad. Readings shall be taken every 1/2 minute after final set, and cracking time shall be recorded as the total elapsed time from final set to when an audible snap is heard upon bending the patty of mixed paste with a quick bend.

4.4.7 Compressive strength. Compressive strength of the mixed paste shall be tested in accordance with American Dental Association Specification No. 30.

4.5 Inspection of packaging. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING.

5.1 Preservation. Peservation shall be level A or C as specified (see 6.1).

5.1.1 Level A.

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5.1.1.1 Unit package. Each unit consisting of one tube of base paste, one tube of accelerator paste, one pad of paper, and instructions for mixing and use shall be packaged in a box of appropriate size and design conforming to PPP-B-566 or PPP-B-676. The box design shall include suitable interior fittings or cushioning to prevent damage to the contents. Closure shall be as specified in the box specification.

5.1.1.2 Intermediate package. Thirty-six units shall be packaged in a box conforming to PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.1.2 Level C. The level C preservation shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Four intermediate packages shall be packed in an exterior container designed for a type 2 load conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Four intermediate packages shall be packed in an exterior container designed for a type 2 load conforming to PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the box specification.

5.2.3 Level C. The level C packing shall conform to MIL-STD-794 requirements for this level.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Levels A and B. Pulp protectors packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets conforming to MIL-STD-147.

5.2.4.2 Level C. Pulp protectors packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

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5.3.1 Levels A, B, and C. Each unit package, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129. Marking shall include the lot or control number and the legend: "Store at controlled room temperature (59°-86°F)," or similar commercial legends. Date of manufacture shall be shown in lieu of date packed.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following item appearing in the Federal Supply Catalog.

National Stock Number	Item Identification
6520-00-074-4193	PULP PROTECTOR, DENTAL, Zinc Oxide and Eugenol, Hard Setting

6.3 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

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MILITARY INTERESTS:

Custodians:
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Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DoD-MB

Agent:

DLA - DM

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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